

Georg von Westphalen

MEDKAYFAB PHYSIOLOGY WRESTLING: LIFE-FORCE LOGIC & HOMEOSTATIC HEAVYWEIGHTS

CALL DOC
FRIZZLE!

PROTECT THE SET-POINT.
MASTER THE LOOP.
STAY FLUFFY!

MEDKAYFABE!



60mV
50mV
40mV
30mV
20mV
10mV
0mV
-10mV
-20mV
-30mV
-40mV
-50mV
-60mV
-70mV
-80mV
-90mV

ATPase-the-Vault-Keeper (Na⁺/K⁺-Pump)

STAT: Consumes 30% of ATP
3 Na⁺ Out, 2 K⁺ In



POWER: The Foundation-Charge. Creates the Potential-Energy-Reservoir.
LORE: Priary active transport. Maintains concentration gradient to power secondary moves.

Aqua-the-Flash-Flood (Aquaporins)

STAT: High permeability to H₂O
/ ADH-dependent



POWER: The Osmotic-Rush. Opens water-gates to equalize Concentration-Stats.
LORE: Facilitated diffusion of water. Responds to osmotic pressure.

Nernst-the-Potential-Judge (Equilibrium Potential)

STAT: $E = (61/z) \times \log\left(\frac{[out]}{[in]}\right)$



POWER: The Voltage-Verdict. Defines the precise Balance-Point for a single ion.
LORE: Defines the Target-Voltage that each ion strives to reach when its gate opens.

Resto-the-Steady-State (Resting Membrane Potential)

STAT: -70mV to -90mV
/ Driven by K⁺ leak



POWER: The Ready-Stance. Keeps the Cell-Engine idling, poised to strike.
LORE: Weighted average of all conductances, dominated by E-Leak-Channels.

KayfaBoGGLE!

KayfaBoGGLE!

$$E_n = \frac{61}{z} \times \frac{\log(out)}{\log(in)}$$

$$\frac{\log(in)}{\log(out)} \times \log \frac{a}{z} = \dots$$

What the FLUFF?!

Depolo-the-Rising-Strike (Depolarization)

STAT: Threshold -55mV
Rapid Na^+ influx



POWER: The Signal-Ignition. Triggers a massive Voltage-Explosion.
LORE: Positive feedback loop. Opening voltage-gated Na^+ channels flips the charge to positive.

Repolo-the-Cool-Down (Repolarization)

STAT: K^+ efflux
Inactivation of Na^+ channels



POWER: The Reset-Wave. Drags the charge back to Resting-State.
LORE: Delayed-rectifier K^+ channels unlock the cell so it can fire again.

Myelin-the-Speed-Booster (Saltatory Conduction)

STAT: Velocity $10\times$
Nodes of Ranvier



POWER: The Jump-Move. Forces the Electric-Strike to jump gaps at lightning speed.
LORE: Increases membrane resistance, decreases capacitance.

Calcium-the-Bridge-Linker (Excitation-Contraction)

STAT: SR Released
Binds Troponin C



POWER: The Squeeze-Trigger. Unlocks the Actin-Myosin-Handshake.
LORE: Removes Tropomyosin-Shield so Myosin-Power-Stroke can begin.

Starling-the-Elastic-Law (Length-Tension)

STAT: SV increases
with EDV



POWER: The Snap-Back. The more you stretch the Heart-Engine, the harder it snaps shut.
LORE: Increased stretch optimizes Actin-Myosin overlap for stronger contraction.

Pre-Load-the-Tank (End-Diastolic Volume)

STAT: Venous Return / CVP



POWER: The Fuel-Pressure. Determines Blood-Fuel in the engine before the squeeze.
LORE: End-diastolic filling volume driven by venous tone and total Fluid-Volume.

Pacem-the-Steady-Drummer (SA Node)

STAT: Phase 4
Funny Currents /
Automaticity



POWER: The Rhythm-Master. Fires a constant Electric-Pulse to dictate match speed.
LORE: Spontaneous depolarization via slow Na^+ leak until Fire-Threshold is reached.

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After-Load-the-Wall (Systemic Resistance)

STAT: MAP / TPR



POWER: The Exit-Resistance. The heavy door the heart must push against.
LORE: Pressure overcome to eject blood via vasoconstriction.

Baro-the-Pressure-Sentry (Baroreceptors)

STAT: Carotid Sinus
Negative Feedback



POWER: The Quick-Stabilizer. Sends Emergency-Signal to Clench-the-Vessels and Speed-the-Heart.
LORE: Decreased stretch = decreased firing = increased Sympathetic output (Fight-Move).

Compli-the-Stretchy-Vessel (Compliance)

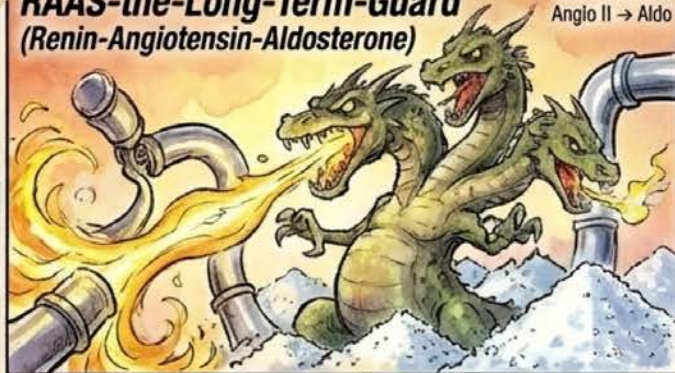
STAT: Change in V / Change in P
(High in veins)



POWER: The Pressure-Buffer. Holds massive blood volumes without Pressure spikes.
LORE: Veins act as Volume-Buffers; Arteries act as Pressure-Pipes.

RAAS-the-Long-Term-Guard (Renin-Angiotensin-Aldosterone)

STAT: Renin →
Angio II → Aldo



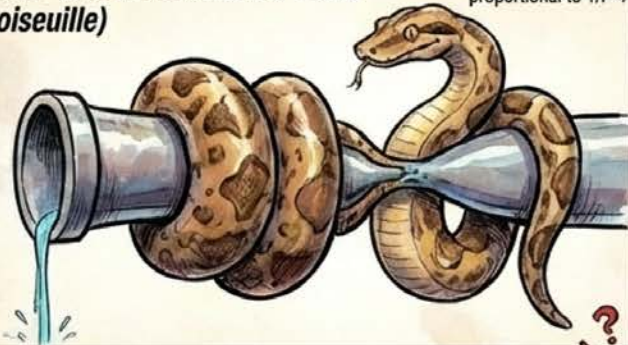
POWER: The Volume-Restorer. A Hormone-Symphony to save salt/water and Clench-the-Pipes.
LORE: Angio II is the Vessel-Squeezer; Aldo is the Salt-Hoarder.

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Flow-the-Resistance-Law (Poiseuille)

STAT: Resistance
proportional to $1/r^4$



POWER: The Radius-Rule. A tiny squeeze causes a massive Resistance-Jump.
LORE: Doubling radius increases flow 16-fold. Radius is the Ultimate-Lever.

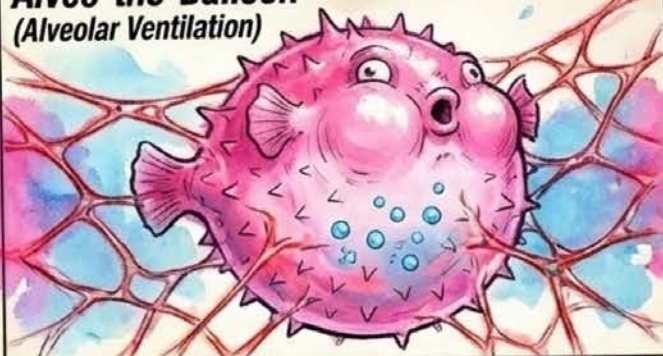
Diagnosis?!



GW Georg v. Westphalen

Alveo-the-Balloon (Alveolar Ventilation)

STAT: $VA = (TV - DS) \times RR$



POWER: The Fresh-Air-Swap. Ensures Fresh-Oxygen-Tokens reach the Exchange-Surface.
LORE: Effective ventilation subtracts Dead-Space air from each breath.

Fick-the-Diffusion-Master (Fick's Law)

STAT: Flux proportional to
(Area x Pressure) / Thickness



POWER: The Gas-Bridge. Determines how fast gases Jump-the-Gap.
LORE: The Oxygen-Jump fails if Membrane-Wall gets too thick (Fibrosis).

Surfact-the-Anti-Collapse (Surfactant)

STAT: Type II
Pneumocytes



POWER: The Balloon-Support. Biological-Lubricant prevents tiny balloons from snapping shut.
LORE: Law of Laplace. Prevents small alveoli from collapsing into large ones.

Bohr-the-Shift-Strategist (Bohr Effect)

STAT: H^+ , CO_2 , Temp
→ Right Shift



POWER: The Oxygen-Unloader. Forces Hemoglobin-Trucks to drop oxygen in hot, acidic arenas.
LORE: Acidosis reduces Hb grip, delivering O_2 to working muscles.

GFR-the-Clearance-King (Glomerular Filtration Rate)

STAT: 125 mL/min (Inulin Gold Standard)



POWER: The Trash-Filter. Determines Flow-Rate of the Waste-Stream.
LORE: Driven by Hydrostatic-Pressure pushing fluid through Fenestrated-Slits.

Loop-the-Concentrator (Loop of Henle)

STAT: Countercurrent Multiplier (1200 mOsm)



POWER: The Salt-Trap. Builds a massive Salty-Desert to pull water out later.
LORE: Ascending/descending permeability rules build an osmotic gradient.

Prox-the-Reabsorber (Proximal Tubule)

STAT: 65% Na⁺, 100% Glucose Reabsorbed



POWER: The Resource-Saver. Snatches Gold-Tokens back before they hit the Urine-Exit.
LORE: Mass-bulk reabsorption driven by Sodium-Cotransport-Moves.

Aldo-the-Salt-Hoarder (Aldosterone)

STAT: Na⁺ In, K⁺/H⁺ Out (Distal Tubule)



POWER: The Volume-Lock. Commands Distal-Pipes to drag water back via salt.
LORE: Increases ENaC-Channels to boost sodium reabsorption and blood volume.

Bicarb-the-Buffer-Shield (HCO_3^-)

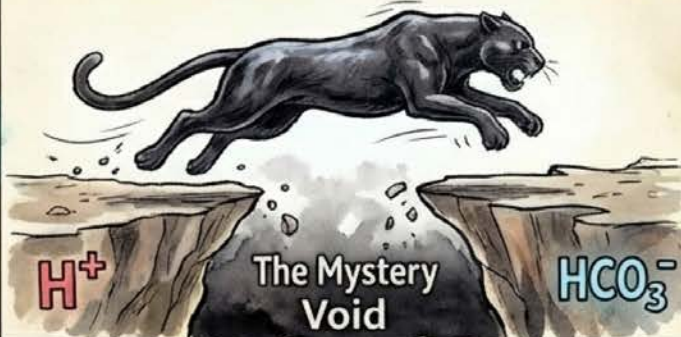
STAT: 22-26 mEq/L



POWER: The Acid-Soaker. Catches Proton-Acid-Darts before they burn the Enzyme-Gears.
LORE: Henderson-Hasselbalch system working with lungs and kidneys to keep pH 7.4.

Anion-Gap-the-Mystery-Math

STAT: $\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$
/ Normal 8-12



POWER: The Poison-Detector. Spots a Hidden-Acid-Saboteur in the ring.
LORE: Law of Electroneutrality; represents unmeasured anions balancing the charge.

CO_2 -the-Respiratory-Acid (PaCO_2)

STAT: 35-45 mmHg



POWER: The Rapid-Acidifier. If Lung-Balloons stall, CO_2 crashes the pH in minutes.
LORE: $\text{CO}_2 + \text{H}_2\text{O} \leftrightarrow \text{H}_2\text{CO}_3$. CO_2 is a volatile acid.

Renal-the-Acid-Excreter (Renal Compensation)

STAT: Secretes H^+ /
Regenerates HCO_3^-



POWER: The Slow-Cleanup. The Long-Match to pee out acid.
LORE: Metabolic compensation takes 2-3 days to counter-move a respiratory attack.

Insu-the-Storage-Master (Insulin)

STAT: Tyrosine Kinase Receptor



POWER: The Sugar-Vault. Opens GLUT-Gates to move Sugar-Tokens into storage.
LORE: Fed-state commander; promotes lipogenesis and glycogen synthesis.

Corti-the-Stress-Shield (Cortisol)

STAT: Glucocorticoid / Diurnal AM peak



POWER: The Long-Term-Survival. Prepares for a Long-Siege, blunting the Immune-Riot.
LORE: Prioritizes current survival over future repair (causes tissue wasting).

Gluca-the-Sugar-Burner (Glucagon)

STAT: Alpha-cells / GPCR



POWER: The Energy-Release. Raids the Liver-Vault to flood blood with Energy-Tokens.
LORE: Fast-state commander; the eternal nemesis to Insu.

Theo-the-Thyroid-Throttle (T3/T4)

STAT: Na⁺/K⁺ ATPase synthesis



POWER: The Metabolic-Speed. Determines how fast Cell-Engines run.
LORE: Revs up system metabolism by increasing Vault-Keepers and Beta-Receptors.

Gastri-the-Acid-Igniter (Gastrin)

STAT: G-cells /
Parietal stimulation



POWER: The Fire-Starters. Commands Stomach-Pits to pour out Acid-Fire.
LORE: Activates Proton-Pump-Engine (H^+/K^+ ATPase).

Secreti-the-Bicarb-Messenger (Secretin)

STAT: Responds to
Duodenum Acid



POWER: The Fire-Extinguisher. Tells Pancreas-Factory to spray Bicarb-Shields.
LORE: Nature's antacid; protects intestinal wallpaper from Gastri's fire.

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CCK-the-Gallbladder-Squeezer (Cholecystokinin)

STAT: Responds
to Fat/Protein



POWER: The Bile-Release. Squeezes Gallbladder-Sponge to spray Bils-Detergent.
LORE: Slows stomach emptying so the small intestine isn't overwhelmed.

Intrinsic-the-B12-Escort (Intrinsic Factor)

STAT: Parietal cells
/ Binds B12



POWER: The Vitamin-Bodyguard. Protects B12 from Acid-Vandals.
LORE: Lack of escort destroys B12, causing Macro-Max-Anemia and nerve melt.



KayfaBINGO!



Erythro-the-O2-Truck (RBC)

STAT: 120-day lifespan
/ No nucleus



POWER: The Oxygen-Courier. Folds and squeezes to carry Oxygen-Crates.
LORE: Navigates narrow Capillary-Alleys to supply the Muscle-Arena.

Coag-the-Fibrin-Web (Secondary Hemostasis)

STAT: Intrinsic/Extrinsic
→ Factor X



POWER: The Permanent-Seal. Turns the Temporary-Plug into solid Armor-Clot.
LORE: Thrombin converts Fibrinogen to a solid Fibrin web.

Hemostat-the-Leak-Stopper (Primary Hemostasis)

STAT: Adhesion
→ Activation
→ Aggregation



POWER: The Temporary-Plug. Throws Platelet-Bricks into the hole.
LORE: vWF is the glue linking bricks to the damaged collagen walls.

Plasm-the-Clot-Dissolver (Fibrinolysis)

STAT: Plasminogen
→ Plasmin / tPA



POWER: The Arena-Cleanup. Eats the web so the Blood-Flow-Highway is clear.
LORE: Digests fibrin. Too slow = Clot-Monster (Thrombosis) blocks the road.

What the
FLUFF?!

Proprio-the-Position-Sense (Mechanoreceptors)

STAT: Muscle Spindles
/ Golgi Tendon



POWER: The Invisible-Map. Tells Brain-Bunker exactly where limbs are in space.
LORE: Essential for Balance-and-Coordination-Moves.

Noci-the-Pain-Alarm (Nociceptors)

STAT: A-delta (Fast)
vs C-fibers (Slow)



POWER: The Damage-Signal. Screams FIRE! when Arena-Walls are crushed.
LORE: Transforms tissue-trauma into electrical-panic.

Baro-the-Static-Balance (Vestibular System)

STAT: Semicircular canals
→ Otoliths



POWER: The Gravity-Anchor. Detects if the Arena is tilting or spinning.
LORE: Fluid moving over hair-cells detects rotation and tilt.

Autono-the-Balance-Beam (ANS)

STAT: Sympathetic vs
Parasympathetic



POWER: The Arena-State-Selector. Selects Combat-Mode vs Recovery-Mode.
LORE: Antagonistic balancing of involuntary systems to match environment stress.

Diagnosis?!

GnRH-the-Pulsatile-Boss (Hypothalamus)

STAT: Frequency
dictates LH/FSH



POWER: The Master-Drummer. Tells Pituitary-Guards when to release hormones.
LORE: If the pulse is steady instead of rhythmic, the lifecycle engine stalls.

FSH-the-Follicle-Farmer (Follicle Stimulating Hormone)

STAT: Ovarian /
Sertoli
stimulation



POWER: The Growth-Nurturer. Waters the seeds for Follicle-Knights.
LORE: Helps eggs and sperm mature for the biological match.

LH-the-Ovulation-Trigger (Luteinizing Hormone)

STAT: The
Mid-cycle Surge



POWER: The Egg-Launch. Forces the Egg-Knight to burst out.
LORE: The mid-cycle surge forces ovulation; in males, stimulates Testo.

Oxyto-the-Squeeze-Hormone (Oxytocin)

STAT: Posterior
Pituitary /
Ferguson Reflex



POWER: The Birth-Push. Causes uterine walls to clench and milk letdown.
LORE: Links physical squeezing-action to social bonding.



Hypothal-the-Thermostat (Hypothalamus)

STAT: Set-point 37.0°C



POWER: The Temp-Governor. Monitors Blood-Temperature to trigger Shakes or Floods.
LORE: Pyrogen-the-Fire-Starter triggers Fever-Move by turning up the dial.

Shiver-the-Heat-Generator (Involuntary Muscle)

STAT: 400% increase
in heat



POWER: The Emergency-Warmth. Forces muscles to twitch for friction-heat.
LORE: Inefficient for movement, highly efficient for thermal energy.

Stay fluffy!

Sweat-the-Evaporator (Eccrine Glands)

STAT: Latent heat
of vaporization



POWER: The Liquid-Coolant. Evaporates Heat-Tokens into the air.
LORE: The most effective cooling move; fails if humidity-stat is too high.

Vasocon-the-Heat-Saver (Sympathetic alpha-1)

STAT: Alpha-1 activation
in skin



POWER: The Peripheral-Shutdown. Clamps surface-pipes to prevent radiative heat loss.
LORE: Keeps Warm-Blood-Tokens safe at the price of icy hands.

The Physiological Attending (Doc FrizzleBob)

STAT: Master of the Loop



LORE: Respect the set-point! Check pressure, check pH, and follow the flow!
POWER: The System-Stabilizer.

Quest 1: The Mountain-Climb

LORE: 5000m altitude. Low O₂.



QUEST: Link Bohr-the-Shift (drops O₂ to tissues) and Renal-the-Acid-Excreter (compensates for hyperventilation blow-off).

Quest 2: The Desert-Run

LORE: Fluid-loss, dropping BP.



QUEST: Trigger the RAAS-Dragon to squeeze pipes and Aldo-the-Armadillo to lock down the salt/water vault.

Quest 3: The Sugar-Crash

LORE: 24-hr fast. Sugar-Stat 60.



QUEST: Gluca-the-Fox raids the liver-vault; Corti-the-Turtle blunts tissue repair to survive the siege.

Call Doc Frizzle! MedKayfabe!